

Math Test

Multiple Choice Questions:

1x10 = 10

1. What is the formula for compound interest?

- A) $SI = (P \times R \times T) / 100$ B) $CI = P(1 + R/100)^n - P$ C) $CI = P \times R \times T$ D) $CI = P + R + T$

2. Find the compound interest on ₹1000 at 10% per annum for 2 years.

- A) ₹200 B) ₹210 C) ₹220 D) ₹240

3. The difference between compound interest and simple interest on a sum for 2 years at 10% is ₹50. Find the principal.

- A) ₹4000 B) ₹5000 C) ₹6000 D) ₹7000

4. If interest is compounded half-yearly, the rate becomes:

- A) Double B) Half C) Same D) Triple

5. If a sum doubles in 5 years at compound interest, how much will it be in 10 years?

- A) 2 times B) 3 times C) 4 times D) 5 times

6. $a^3 + b^3 = ?$

- A) $(a + b)(a^2 + ab + b^2)$ B) $(a - b)(a^2 + ab + b^2)$ C) $(a + b)(a^2 - ab + b^2)$ D) $(a + b)(a^2 + ab - b^2)$

7. $a^2 + b^2 = ?$

- A) $(a + b)(a - b)$ B) $a^2 + 2ab + b^2$ C) $(a + b)^2 + 2ab$ D) None of these

8. $(a + b)^3 = ?$

- A) $a^3 + b^3 + 3ab(a + b)$ B) $a^3 + 3a^2b + 3ab^2 + b^3$ C) None of these D) Both A & B

9) If $(a + b + c) = 0$ then which is correct?

- A) $a^3 + b^3 + c^3 = 3abc$ B) $a^2 + b^2 + c^2 = 2abc$ C) $(a + b)^3 = a^3 + b^3$ D) None of these

10) $(2x-3y)(2x+3y) = ?$

- A) $4x^2 + 9y^2$ B) $4x^2 - 9y^2$ C) $(4x + 9y)^2$ D) $(4x - 9y)^2$

Solve the Following:

2 x 5 = 10

1. A man invests ₹5000 for three years at a certain rate of interest, compounded annually. At the end of one year, it amounts to ₹5600. Calculate the rate of interest per annum.

2. Find the amount and the compound interest on ₹8000 for 4 years at 10% per annum, interest reckoned yearly.

3. The present population of a town is 10000. If the population increases at the rate of 15% every year, what will be the increase in the population in next 2 years?

4. Simplify: $(9a + 5b)^2 - (9a - 5b)^2 - 180ab$

5. Simplify: $(5x - 3y + 3)(5x + 3y + 3)$

Solve the Given Questions:

5 x 2 = 10

1. i) Find the product of $(x - 3)(x + 7)(x - 4)$ and also find the coefficient of x^2 and x . **3 Marks**

ii) Using Suitable identity, find the value of: **2 Marks**

$$6.8 \times 6.8 \times 6.8 + 3.4 \times 3.4 \times 3.4$$

$$6.8 \times 6.8 - 6.8 \times 3.4 + 3.4 \times 3.4$$

2) At what rate percent p.a. compound interest would ₹80000 amount to ₹8820 in two years, interest being compounded yearly. Also find the amount after 3 years at the above rate of compound interest.